

Xinyu (Brian) Guo

(314) 680-8961 | xinyug@usc.edu | <https://www.xinyuguo.com>

PROFESSIONAL SUMMARY

AI for Science Researcher (Ph.D. Candidate, USC) specializing in the development of **Geometric Representation Learning** and **Biological Foundation Models**. Proven track record in designing state-of-the-art architectures (GNNs, SSMS/Mamba) for complex biological sequences and spatial data, achieving up to 68% memory reduction for long-horizon tasks. Expert in scaling distributed training on GPU clusters and bridging multi-omics data with interpretable deep learning.

TECHNICAL EXPERTISE

AI/ML Architectures: Foundation Models (DNA/RNA/LLM), GNNs/GAT, Transformers, SSMS/Mamba, Self-Supervised Learning (DINO/BYOL).

Scientific Domain: Spatial Transcriptomics, Multi-omics pipelines, GWAS/TWAS, Biological Foundation Models.

Systems & Scaling: PyTorch, PyTorch Geometric, Distributed Training (DDP/FSDP), CUDA, Model Quantization.

Languages: Python, R, C++, SQL, JavaScript/TypeScript.

SELECTED RESEARCH EXPERIENCES

Biological Foundation Models & Statistical Learning | Lead Researcher | 2022 – Present

University of Southern California, Advisor: Dr. Liang Chen

- Developed a self-supervised contrastive learning method with GNN/GAT architectures to analyze spatial transcriptomics data, enabling integration with large-scale multi-omics datasets and genetic risk evaluation.
- Proposed a fusion architecture utilizing weighted message-passing GNNs and Attention mechanisms to project GWAS/TWAS signals into spatial embeddings.
- Developed a cross-modality foundation model integrating DNA sequence, RNA expression, and Spatial Transcriptomics (ST) to identify tissue-specific disease risk regions. Focused on AI model explainability to enable interpretable disease microenvironment analysis, mapping genetic variants to specific spatial risk regions.
- Benchmarked large DNA models (AlphaGenome, Enigma) to predict spatial variant-to-gene perturbation effects.
- Comparatively studied methods for identifying cell cycle phases from single-cell RNA-seq data.

Statistical Genetics & Transcriptome Association Studies | Lab Member | 2020 – 2022

Johns Hopkins University, Advisors: Dr. Nilanjan Chatterjee & Dr. Diptavo Dutta

- Developed a cross-tissue subset-based meta-analysis method (CSTWAS) to identify potential "active" tissues with improved power, published in *Human Genetics and Genomics Advances*.
- Performed transcriptome-wide and single-cell level association studies to identify genes linked to renal cell carcinoma, with results published in *The American Journal of Human Genetics*.
- Modeled genome-wide association study (GWAS) results with single-cell ATAC-seq data to interpret the etiology for human diseases at the cellular level.

Multi-Omics Pipeline Development & Cancer Resistance | Lab Member | 2021 – 2022

Johns Hopkins University, Advisors: Dr. Hongkai Ji & Dr. Eneda Toska

- Built a comprehensive RNA-seq, ChIP-seq, and ATAC-seq analysis pipeline enabling robust differential analysis, gene set enrichment, and motif discovery.
- Elucidated how transcriptional and epigenetic regulators mediate hormone-driven cancer tumor resistance to therapeutics.
- Contributed to findings published in *Cell Reports*, *Oncogene*, and *Cancer Research* regarding chromatin modifier KMT2D and therapeutic responses in breast and prostate cancer.

Predictive Clinical Informatics & EHR Implementation | Research Assistant | 2020

Washington University in St. Louis, Advisor: Dr. Andrew Michelson

- Built Lasso and Logistic Regression models to predict respiratory failure in COVID-19 patients.
- Successfully implemented predictive models in the Barnes-Jewish Hospital EHR system to enable early identification of high-risk patients and timely ICU resource deployment.

LLM Foundation Training from Scratch | *Research Project* | 2025

- Engineered and trained a 560M parameter decoder-only LLM from scratch using BPE tokenization and distributed PyTorch (DDP) across 8×A100 80G GPUs.
- Managed the full model training lifecycle, including mixed-precision training and optimized gradient synchronization to maximize cluster throughput.

EDUCATION

Ph.D., Computational Biology and Bioinformatics

University of Southern California | GPA: 3.95/4.0 | Aug 2022 – Present

- Andrew J. Viterbi Fellowship for outstanding Ph.D. students in computational sciences

M.S., Biostatistics

Johns Hopkins University | GPA: 3.97/4.0 | Aug 2020 – May 2022

- Delta Omega Honorary Society in Public Health

B.A., Mathematics and Computer Science

Washington University in St. Louis | GPA: 3.89/4.0 | Aug 2018 – May 2020

- Cum Laude, Dean's List

SELECTED PUBLICATIONS

- Guo, X., Chen, L. From G1 to M: a comparative study of methods for identifying cell cycle phases. *Briefings in Bioinformatics*.
- Bianchi, A., ..., Guo, X., ..., Merchant, N. Cell-autonomous Cxcl1 sustains tolerogenic circuitries and stromal inflammation via neutrophil-derived TNF in pancreatic cancer. *Cancer Discovery*.
- Dutta, D., Guo, X., et al. Transcriptome-and proteome-wide association studies identify genes associated with renal cell carcinoma. *The American Journal of Human Genetics*.
- Guo, X., Chatterjee, N., Dutta, D. Subset-based method for cross-tissue transcriptome-wide association studies improves power and interpretability. *Human Genetics and Genomics Advances*.